

An End-to-End Agent for Kaggle Competitions

Benchmarked against two frozen yardsticks on 20 competitions

Report v5 (August 2026)

All printed numbers are real measurements as of 2026-08-17; TBD = awaiting the leaderboard scoring step.

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Abstract

We present **my-agent**, an end-to-end agent for machine-learning competitions, and the design decision at its centre: *where external knowledge enters the pipeline*. The agent runs the full loop — problem identification, EDA, feature engineering, modelling, evaluation, submission — with an LLM reasoning at every stage, Auto-ML libraries invoked through the shell, a cross-competition experience library supplying evidence-backed priors, and a tree search over candidate configurations as the optimisation loop. Measuring our own first design, we found knowledge injected downstream at the blending stage worth only $\sim 10^{-5}$; the architecture therefore injects it upstream through a *problem-identification layer* that classifies task family, train–test window relation, split policy, and external-data need before any modelling, emitting typed operators into a whitelisted, leakage-guarded external-data module. Held to real-leaderboard discipline, the layer separates cleanly (it fires on exactly the 2 of 18 baseline competitions that need external data, silent on the other 16 — the silence being the informative half, since the two positives are partly in-sample); in matched-configuration designs across three firing-class competitions all six external-data arms beat their own baselines with paired significance; and on the benchmark’s canonical failure a with-layer versus without-layer ablation passes the paired test — while no local signal we tested can select the *form* the injection should take, a negative result sharp enough that the one selector surviving hindsight was killed by its first pre-registered prediction. To place the agent, we benchmark it against two frozen, methodologically distinct yardsticks — **AIDE**, LLM tree search, and the **NVIDIA Kaggle Agent**, which reproduces the highest-voted public kernel — on 20 Kaggle competitions under a mechanically enforced fairness protocol, with my-agent’s lanes running in credential-free sandboxes and audited from their own transcripts, plus three mid-complexity competitions spanning NLP, physiological time series, and medical imaging, where my-agent takes the three-way best on all three. The benchmark lanes’ submissions exist, but the leaderboard scoring step — a separate credentialed operator step — has not yet run, so my-agent’s standings against the yardsticks are TBD; once scored, every duel is adjudicated with an error model built from Kaggle’s own public/private split and from measured run-to-run agent variance. That apparatus is not optional: in this study’s own development runs, local cross-validation contradicted the leaderboard in 9 of 13 competitions and the ranking’s winner flipped when the composition of the evaluation set changed — which is why an agent’s capability claim, ours included, must be stated together with the error model and the set it was measured on.

1 Introduction

Machine-learning competitions are *scorable tasks*: a submission receives a single measurable quality score on a hidden test set, and a public leaderboard records how it compares against thousands of human teams. Solving them end-to-end — data understanding, validation design, feature engineering, model

selection, ensembling — has traditionally demanded expert labor and extensive trial and error. LLM-driven coding agents have recently begun to automate this loop: AIDE performs LLM-guided tree search over candidate solutions [2]; ERA couples an LLM rewriter with tree search and externally injected research ideas, outperforming strong baselines across 16 Kaggle Playground competitions [1]; and industrial agents skip search altogether by reproducing the strongest public solution to a given competition [3].

This report is about one such agent, **my-agent**, built over the course of this internship: a hybrid design in which an LLM reasons at every stage of the competition pipeline, Auto-ML libraries are invoked through the shell for what they do best, a cross-competition experience library supplies evidence-backed priors, and a tree search over candidate configurations serves as the optimisation loop (Figures 1 and 2).

The design question we ended up caring most about is *where external knowledge should enter the pipeline*. Our first answer was the conventional one — inject it downstream, among the candidates being blended — and measuring our own implementation showed that answer to be worth about $\sim 10^{-5}$. Two development findings pointed at the fix. First, in development runs local cross-validation contradicted the real leaderboard in 9 of 13 comparable competitions (Section 5.1), so nothing local can be trusted to arbitrate architectural claims. Second, a failure-mode analysis of all three agents studied here (Section 5.2) located their shared gap at problem-level judgment — “does this task need external data?”, “is the test set a future time window?”. The architecture therefore injects knowledge upstream, through a *problem-identification layer*: a dossier stage that runs before EDA, classifies the task, and emits typed operators into a whitelisted, leakage-guarded external-data module (Section 2.2). That layer is evaluated under real-leaderboard discipline in Sections 6.1 and 6.2 — the scored evidence at the centre of this report.

Testing an architectural claim of this kind requires reference points we did not build and cannot tune, so all results are stated against two frozen, methodologically distinct yardsticks: **AIDE**, LLM tree search with no cross-competition memory, and the **NVIDIA Kaggle Agent**, which reproduces the highest-voted public kernel. All three run on 20 Kaggle competitions plus three mid-complexity competitions, and every yardstick submission is scored on the real leaderboard via late submission rather than on any local estimate. **my-agent**’s 20 benchmark lanes ran under the frozen architecture in credential-free sandboxes (Section 3.1); their submissions exist, but the scoring step is a separate credentialed operator step that has not yet run, so every **my-agent** leaderboard cell and paired verdict in the benchmark tables is TBD (Section 4.2).

Doing this honestly turned out to require apparatus of its own (Section 3), because agent comparisons are ordinarily reported as one run and one score per competition with the higher number declared the winner, and two error sources make that fragile: competition metrics carry test-set sampling error, with top-two gaps often at the fourth or fifth decimal place, and agent execution is itself stochastic. We therefore enforce isolation mechanically, adjudicate every verdict with a paired significance test built from Kaggle’s own public/private split, and measure run-to-run variance directly.

What the report contains: the architecture and its injection layer (Section 2.2); the benchmark apparatus and the error model that will adjudicate its TBD standings (Section 3); the 20-competition benchmark with **my-agent**’s local numbers and the yardsticks’ frozen leaderboard scores (Section 4.2); the development findings that motivated the layer (Sections 5.1 and 5.2); the injection layer’s controlled, leaderboard-scored experiments (Sections 6.1 and 6.2); and a mid-complexity three-way across NLP, physiological time series, and medical imaging where **my-agent** takes the best score on all three (Section 7).

2 Methods

2.1 **my-agent**: A Hybrid Kaggle Agent

The method is implemented as a Claude Code Skill (`kaggle-agent`). Its core design is a division of labor between LLM reasoning and Auto-ML tools: the LLM (Claude Code) handles problem understanding, validation-strategy design, creative feature engineering, result interpretation, and deciding the next step; LightGBM / XGBoost / CatBoost / Optuna — invoked by the LLM via the shell — handle model selection, hyperparameter optimization, and stacking/ensembling.

Execution proceeds through the pipeline of Figure 1: data ingestion → **Stage 0.5** problem dossier → EDA → feature engineering → modeling → evaluation → submission. The dossier stage decides what kind of problem the task is before any modeling (Section 2.2); it is evaluated in its own right, under the same paired-test discipline as everything else, in Sections 6.1 and 6.2. Modeling and evaluation run in one of two optimization modes:

- **Linear iteration protocol:** propose a hypothesis, run the experiment, record the result — used for the first pass and for small competitions.
- **Tree search (preferred):** once the linear protocol has produced a baseline single model and at least one blend, the agent switches to tree search. Internal evaluation shows tree search winning 9 of 10 competitions with 1 tie and 0 losses.

Three cross-cutting mechanisms support the pipeline:

Cross-competition experience library (`knowledge/experience.md`): every insight must carry an evidence field (`competition`, `exp #N`, `score A` → `score B`); entries without a score delta are rejected. The agent queries the library before EDA and before modeling, and writes validated new insights back after experiments. This is the most fundamental design difference between my-agent and the two reference methods.

Experiment records. Every experiment’s parameters, features, split scheme, and scores are logged in structured form (`experiments.json`, `experiments_tree_v3.json`), making every number in this report traceable.

External-data layer : realizes the dossier’s typed operators as whitelisted, leakage-guarded external joins, each raced against the no-external baseline inside the search (Section 2.2).

Figure 1 summarizes the architecture and how the three mechanisms connect to the search loop.

Tree-search protocol. Every run in this report uses the tree-search harness of Figure 2 (`tree_search/harness_v3.py`): ensembles are first-class node kinds with out-of-fold caching and weight search, and a budget phase machine exploits until all lineages plateau, then forces one exploration burst, then stops. An earlier downstream variant of the harness injected external ideas at the blend stage; no benchmark submission here used it, and its measured near-null effect is what motivated the upstream layer (Section 2.2).

2.2 The Injection Layer: Dossier, External Data, and the Typed Contract

The blue elements of Figures 1 and 2 form a single layer, and its placement carries the design argument. An earlier downstream variant had injected external ideas at the blend stage and measured a negligible effect (reweighting never displaced the convex-optimum champion in 8/8 rigorous competitions; pool expansion helped 2/8 by $\sim 10^{-5}$); the architecture moves the injection point upstream to problem identification, where the failure-mode analysis (Section 5.2) locates the shared gap of all three agents:

Stage 0.5 — problem dossier. Before EDA, the agent answers “what kind of problem is this?”: task family, train–test window relation, split policy, and external-data need, written to a structured `dossier.json`. Judgments are grounded in `knowledge/task_priors.md`, a library of task-level priors each carrying an evidence field from our own benchmark (e.g., the shuffled-KFold 4.5× bias; the GDP-join CV gap on jan-2022). Reading competition-specific discussions or kernels is forbidden — the dossier must derive from the problem statement plus task-*type* knowledge only, keeping the method distinct from reproduction-based agents. The dossier is a prior, not a conclusion: EDA must verify its hypotheses and record a verdict.

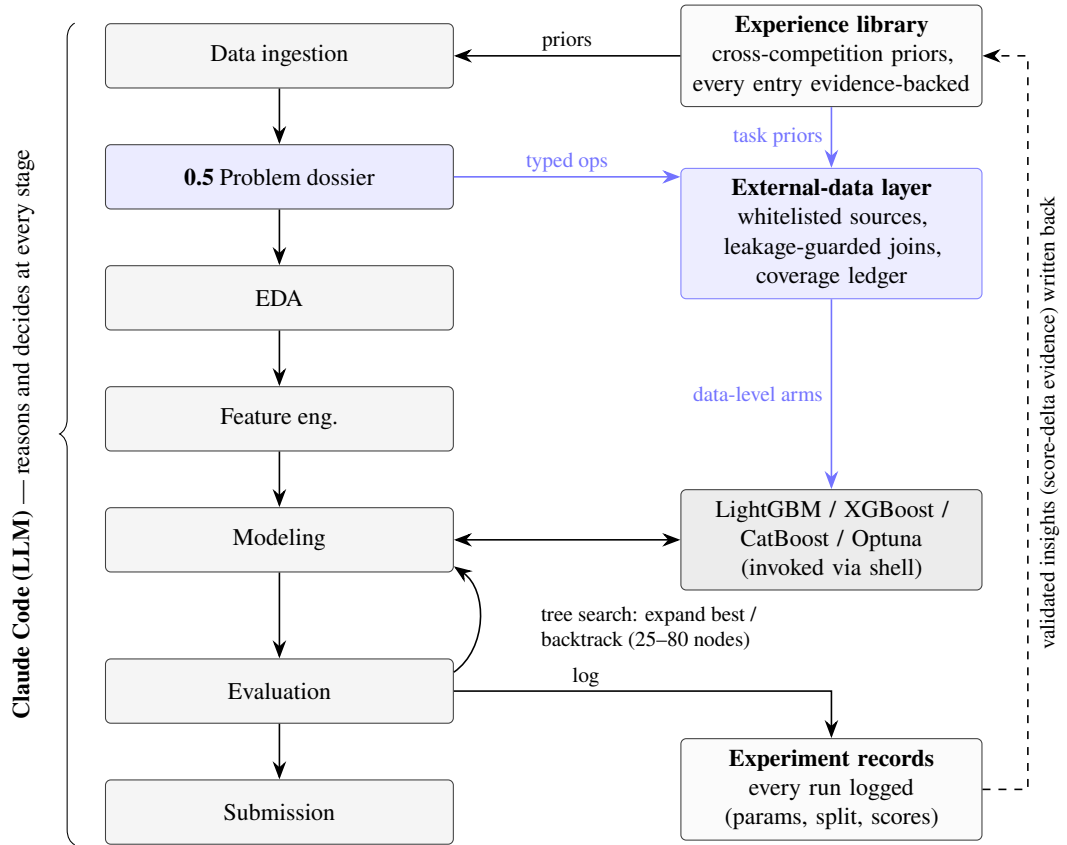


Figure 1: Architecture of my-agent. The LLM drives every stage; modeling and evaluation iterate via tree search, invoking Auto-ML tools through the shell, while the experience library and experiment records close the evidence loop. Blue: the injection layer (Section 2.2) — the Stage 0.5 dossier emits typed operators that drive the external-data layer, whose data-level arms race against the no-external baseline.

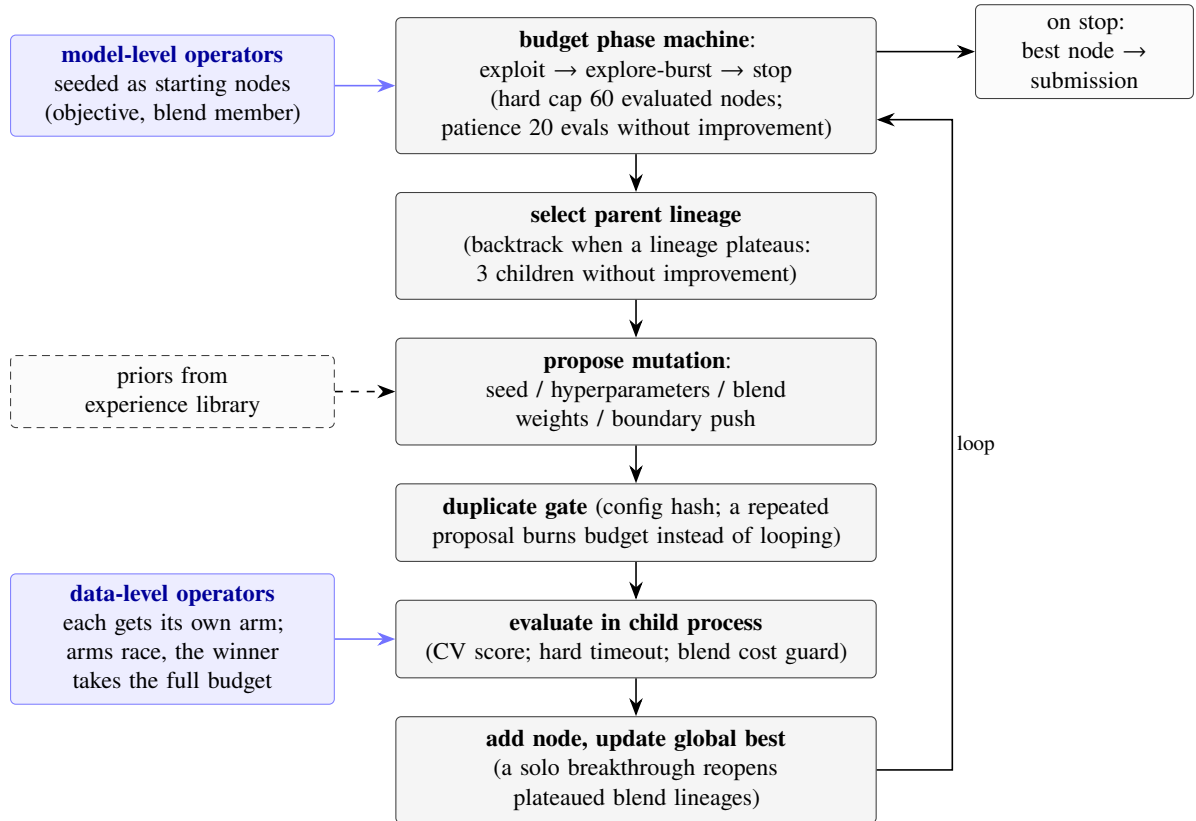


Figure 2: The tree-search loop used in all my-agent runs of this report (22–80 nodes per competition). Each node is a complete, executable solution scored by local cross-validation; lineages are abandoned on plateau, and the budget phase machine forces one exploration burst before stopping. Blue: the injection layer’s operator entry points (Section 2.2) — model-level operators seed starting nodes; data-level operators build separately raced arms.

External-data module. A whitelist of public sources (eleven World Bank indicators, ECB monthly FX, OWID COVID, holiday calendars, ISO tables), each fetched with recorded snapshots and joined through leakage-guarded as-of merges: an external value may reach a prediction row only if dated at or before the row’s period minus a declared lag; violations raise an error; numeric time columns must declare their unit; every join writes an audit log. The module was adversarially verified (21 reproduced findings, including three silent-future-leak classes, all fixed with regression tests).

Search integration. Data-level dossier ideas spawn their own variant workspaces raced at small tree-search budget; model-level ideas enter the main tree as seed lineages. The winning lane receives the full search budget.

A typed contract between the two layers. The first version of this stack failed instructively: the dossier correctly proposed World Bank GDP for s3e19 *as a country-level scale covariate*, and the experience library already held the operational recipe, but the execution layer could express exactly one transformation — appending a column to the feature list. Since a gradient-boosted tree cannot extrapolate a feature beyond its training range, the semantics of “scale covariate” evaporated between the layers with nothing raising an error. The fix is a closed vocabulary both layers read (`knowledge/injection_operators.md`): ten typed operators, each with a parameter schema, the evidence justifying it, and explicit conditions under which it must *not* be used. The dossier may emit only these, each carrying its rationale and source prior; anything the vocabulary cannot express is written to `not_recorded` rather than into prose the execution layer will ignore. The dispatcher writes a coverage ledger — proposed, realized, unrealized-with-reason — so an unimplemented capability leaves a trace instead of resembling success. Two automatic pre-conditions guard the level-covariate operators: proportionality and covariate volatility. Conformance is machine-checked on both sides (Appendix A.5).

2.3 Two Reference Methods (Yardsticks)

AIDE (open-source agent, gen-1): LLM-driven tree search. At each step it writes a complete solution, executes it, reads the error or score, and mutates accordingly. No cross-competition memory — every competition starts from zero; the journal only persists within a single run.

NVIDIA Kaggle Agent (skill): its method is “reproduce the strongest public kernel” — research the competition context, select the highest-voted solution-type kernel, port it faithfully, and retrain on local data.

Key methodological decision. Both reference methods are frozen — no improvements of any kind. During the study we found that NVIDIA’s vote-based kernel selection is a structural weakness (Section 5.2), and the skill itself even ships a score-based selection tool, but we deliberately did not enable it. Rationale: the purpose of this study is to use them as fixed yardsticks for measuring my-agent; improving the yardsticks mid-study would make earlier and later data incomparable, and would quietly turn the benchmark into an optimization of the reference methods.

3 Experimental Apparatus

Two pieces of apparatus are prerequisites for every three-agent comparison and every real-leaderboard verdict in this report, and neither is a finding in its own right: a fairness and execution-discipline protocol that makes a score attributable to a method rather than to an accident of the shared machine, and an error model that says which score differences are readable at all. We state both here, once, so that the experimental sections can report results without re-deriving the conditions under which they mean anything. The offline-graded mid-complexity competitions of Section 7 are the one exception to the error model’s reach, noted there.

3.1 Fairness Protocol and Execution Discipline

The three agents share one machine, so four isolation conditions are defined and mechanically enforced rather than left to convention (Table 1): if the agents could read each other’s files, or run concurrently on a contended machine, no verdict below would be attributable to method. Implementation details and the failure record are in the companion engineering log.

Table 1: The four isolation conditions and what enforces each.

Condition	Rule	Enforcement
Data	Each side reads only its own root, containing files validated against the official Kaggle manifest	<code>build_isolated_roots.py</code>
Workspace	The three output directories are fully separated; each side is scored independently	Directory structure
Instruction	Prompts may carry environment facts and metric thresholds only — no modelling recipes, no other side’s scores, no workaround discovered by another side	Manual protocol + incident log
Resource	Only one lane executes at any moment	<code>lane_lock.sh</code> mutex

A fifth rule governs the experimenter rather than the agents: when the environment is defective, fix the environment rather than hint to the affected side. Fixing removes the confound; hinting compensates one side and silently changes what is being compared.

Benchmark execution discipline. For my-agent’s 20 benchmark lanes the protocol is tightened further. A benchmark manifest (`docs/rerun_manifest.json`) fixes exactly one workspace, one canonical evaluator, and one search driver per competition, after a development audit found workspaces accumulating up to 13 candidate evaluators, which would have made “the agent’s run” ambiguous. Each lane executes in an isolated root whose data files are validated against the official Kaggle manifest, holds *no credentials* (the Kaggle token is stripped from the environment, so a lane can neither consult the leaderboard nor submit), sees only its own workspace (the other nineteen are unreadable), and is denied the answer-key and out-of-fold cache channels an earlier audit found reachable. A *per-lane transcript audit* then reads every tool call the lane made — detectors for network access, raw opens of the shared knowledge library that bypass the query interface’s self-citation exclusion, and results of sibling benchmark competitions appearing without a sanctioned source — and the launcher halts the whole queue after two consecutive audit failures, on the reasoning that two in a row indicts the pipeline rather than one lane. Section 4.3 reports the audit gate exercised in both directions on the actual run.

3.2 The Paired Significance Test

Why an error model is needed at all. In the development-lane leaderboard measurements this apparatus was built on, several top-two gaps sat at the fourth or fifth decimal place (s5e10: 0.00003; s6e2: 0.00000). Do such gaps exceed measurement error? If not, the “verdict” is noise — and a report that declares winners at that scale is reporting its own noise back to itself.

How — the error scale comes from the data itself, not from assumptions. Kaggle randomly splits each test set into two disjoint subsets, public and private, and one submission receives one score on each. These are two independent estimates of the same model, so their difference is that competition’s metric sampling noise.

A failed first attempt, worth recording. We initially used a single agent’s own public–private gap as the threshold, and 8 of 10 competitions came out as ties. The reason is that this gap is mostly *shared* across the three agents — which rows land in which subset shifts everyone together. On s5e10 (Table 2) the common shift is 0.00025 against mutual scatter of only 0.00002: using the common shift as the threshold inflates the true noise more than tenfold — measuring a paired question with an unpaired scale.

The correct construction. Measure how much the *gap between two agents* moves across the two subsets — the common drift cancels in the subtraction. Writing d_{pub} and d_{priv} for the between-agent score difference on the public and private subsets, a verdict requires both:

$$\text{sign}(d_{\text{priv}}) = \text{sign}(d_{\text{pub}}) \quad (\text{the ordering replicates on disjoint samples}) \quad (1)$$

$$|d_{\text{priv}}| > |d_{\text{priv}} - d_{\text{pub}}| \quad (\text{the gap exceeds its own movement}) \quad (2)$$

Statistical properties of the decision rule. Algebraically, condition (2) is equivalent to $d_{\text{pub}}(2d_{\text{priv}} - d_{\text{pub}}) > 0$ and therefore *implies* condition (1); we state both because they diagnose different failures. The rule is a conservative screen rather than a calibrated hypothesis test, but its false-verdict rate under a null model is computable. Suppose two agents are truly equivalent on a competition, so that d_{pub} and d_{priv} are independent zero-mean Gaussian noise with $\sigma_{\text{pub}} = k \sigma_{\text{priv}}$. The probability of a spurious verdict (in either direction) is

$$P_{\text{FP}}(k) = \frac{1}{2} - \frac{\arctan(k/2)}{\pi}, \quad (3)$$

which gives 35.2% for equal variances ($k = 1$) and 25.0% for the 20%/80% public/private splits common in these competitions (variance scales inversely with subset size, so $k \approx 2$); condition (1) alone caps it at 50% distribution-free. This is the rule that adjudicates every leaderboard verdict in this report, including the benchmark’s TBD duels once the scoring step runs.

Table 2: Empirical evidence for common drift (excerpt; development-lane submissions). Drift = private minus public score of the same submission. In seven competitions the mutual scatter is less than half the common shift.

Competition	my-agent	NVIDIA	AIDE	Common shift	Mutual scatter
s5e10	+0.00025	+0.00026	+0.00024	+0.00025	0.00002
s6e2	+0.00149	+0.00160	+0.00152	+0.00154	0.00011
s3e11	+0.00063	+0.00065	+0.00056	+0.00061	0.00009
s3e7	−0.00859	−0.00772	−0.00825	−0.00819	0.00087

The second error term: between-run agent variance. The paired test covers test-set sampling noise only; agent execution is itself stochastic. On the four competitions with the benchmark’s tightest development-lane duels, my-agent ran twice fresh under identical conditions (fresh workspace, knowledge library pinned per competition to the last commit before any information about that competition entered it), and AIDE’s pinned re-run pairs against its frozen baseline. Private-leaderboard spreads:

	my-agent	AIDE	development-lane duel gaps
s4e11	0.00021	0.00027	0.00013–0.00033
s3e16	0.00460	0.00056	0.00142–0.00365
s5e10	0.00000	0.00003	0.00003–0.00012
s6e2	0.00000	0.00020	0.00000–0.00008

Three readings, condensed. First, the variance is substantially a property of the *task*, not the agent: both agents agree s3e16 is the noisy competition by an order of magnitude, so error bars must be per-competition, not one global number. Second, several fourth-decimal development-lane gaps sit at or

below a measured wobble (s5e10’s 0.00003 gap exactly equals AIDE’s wobble there), so the combined error model treats gaps at that scale as undecidable — a revision the TBD benchmark verdicts will inherit. Third, the pipeline’s endpoint is far more stable than its route: my-agent’s two runs took visibly different paths on every competition (a linear-protocol blend versus a 60-node tree search on s4e11; a top-30 equal blend versus a 19-node greedy selection on s3e16) yet landed within 0.0002 everywhere except the task both agents find noisy. NVIDIA’s variance term remains unmeasured (Section 8.5).

4 The Three-Way Benchmark

4.1 Overview of Experimental Design

Evaluation set. 20 tabular competitions — 14 from the Kaggle Playground Series, plus afsis-soil-properties (2014, spectral regression), conway-s-reverse-game-of-life (2014, structured multi-output), cat-in-the-dat (2019, purely categorical features), and tabular-playground aug-2022, jan-2022, and sep-2022 (Table 3). Every competition provides both a public and a private test-subset score — the inclusion criterion required by the significance test (Section 3.2). The three agents also ran 4 competitions for which private scores cannot be obtained (late submission closed, rolling leaderboards); these are excluded. The set grew from a frozen 18-competition baseline to 20 when the two firing-class competitions (s5e1, tps-sep-2022) joined on 2026-07-31 after all three lanes ran them under benchmark conditions — a post-hoc expansion disclosed as a limitation (Section 8.4).

Execution conditions. The three agents ran serially under the mutex of Section 3.1. AIDE: 20 steps, its own default budget (conway stopped at 16 due to the 4-hour cap). NVIDIA: reproduced public kernels per its method. my-agent: the full six-stage pipeline including tree search (22–80 nodes per competition; Appendix A.6), each lane under the execution discipline of Section 3.1.

4.2 Three-Way Capability Comparison (Raw Scores)

Goal and procedure. Measure the absolute performance of the three methods on the same set of competitions. Each side independently produces a submission, scored on Kaggle via late submission for public and private scores and a percentile rank (PR). The AIDE and NVIDIA columns of Table 4 are those frozen leaderboard scores. my-agent’s lanes hold no credentials, so *nothing in its column has touched the leaderboard yet*: each lane’s frozen `submission.csv` exists, and scoring is a separate credentialed operator step (`score_lane_submissions.py`; dry-run by default, real submission behind an explicit flag), after which `build_rerun_three_way.py` places the my-agent column against the yardsticks under the paired test of Section 3.2. Until then, every my-agent leaderboard cell, percentile rank, and paired verdict is TBD; the local CV column reports each lane’s own honest out-of-fold estimate, which Section 5.1 shows must not be read as a leaderboard proxy (s3e19’s pooled OOF SMAPE of 6.71 sits against yardstick leaderboard SMAPEs near 48, re-illustrating the point).

PR is the share of that competition’s final private standings the submission would beat (a would-be rank r among N teams gives $PR = 100(N - r)/N$, against the frozen final leaderboard) — an *absolute* capability reference (“where would this agent have placed had it entered?”), separate from the relative ordering among the three. On s3e19 the yardstick PRs of 41.4 and 40.1 place both far behind typical participants; on s3e16 their 96.4 and 91.4 sit in the top tier, where small mutual gaps carry little practical meaning.

4.3 Execution and Audit of the Benchmark Run

The queue ran 2026-08-14 through 08-17 under operator approval, halting twice — once on an environment fault at a lane boundary (which the launcher distinguishes from a lane result by exit timing) and once on its two-consecutive-audit-failures rule — and had 16 of 20 lanes run, one under audit review, and three queued as this draft compiles.

Table 3: Competition descriptions. Tasks fall into two families: classification (binary, ordinal; 7 competitions) and prediction/regression (incl. time series and multi-target; 13 competitions).

#	Competition	Task type	Train rows $\times$ cols
1	s3e1	Regression	$37,137 \times 10$
2	s3e3	Classification (binary)	$1,677 \times 35$
3	s3e5	Classification (ordinal)	$2,056 \times 13$
4	s3e7	Classification (binary)	$42,100 \times 19$
5	s3e9	Regression	$5,407 \times 10$
6	s3e11	Regression	$360,336 \times 17$
7	s3e14	Regression	$15,289 \times 18$
8	s3e16	Regression	$74,051 \times 10$
9	s3e19	Regression (time series)	$136,950 \times 6$
10	s4e11	Classification (binary)	$140,700 \times 20$
11	s5e10	Regression	$517,754 \times 14$
12	s6e1	Regression	$630,000 \times 13$
13	s6e2	Classification (binary)	$630,000 \times 15$
14	afsis-soil-properties	Regression (multi-target)	$1,157 \times 3,600$
15	cat-in-the-dat	Classification (binary)	$300,000 \times 25$
16	conway-s-reverse-game-of-life	Structured multi-output	$50,000 \times 802$
17	tps-aug-2022	Classification (binary)	$26,570 \times 26$
18	tps-jan-2022	Regression (time series)	$26,298 \times 6$
19	s5e1	Regression (time series)	$230,130 \times 6$
20	tps-sep-2022	Regression (time series)	$70,128 \times 6$

The audit gate was exercised in both directions. The sharp case is **s3e19**: the 08-14 lane opened the raw knowledge library four times and issued a genuine socket-opening command, and the audit discarded it. The clean 08-17 re-run scores *worse* on local CV — pooled OOF SMAPE 6.706 against the tainted attempt’s 5.740 — consistent with the contamination advantage being real, and with the audit pricing it correctly. A benchmark that only ever accepts its own lanes has no evidence its audit does anything; this one discarded a better-looking number and kept the honest one. The other two incidents were resolved by reading transcripts rather than trusting either the lane or the auditor: the network detector fired on an ls of a directory named `kaggle-safe-submit` (a manual scan of all 118 tool calls found zero network commands; accepted under a recorded operator waiver, detector fix queued for after the run, since the run root is baseline-frozen), and on s5e10 the flagged raw library opens all occurred strictly *after* the submission was written and validated, so the score stands — the genuine violation there was a hand-edit of the *shared* knowledge library after finalisation, the channel by which one lane’s outcomes could poison later lanes’ priors; it was surgically reverted before any later lane ran.

4.4 Standing of the Three Agents Under the Paired Test

The benchmark’s headline reading is the full duel matrix: 20 competitions $\times$ 3 pairings = 60 duels, each adjudicated by the rule of Section 3.2. Two of every three duels involve my-agent, whose leaderboard scores are TBD, and the yardsticks’ aggregate win rates fold their my-agent duels in — so no row of Table 5 can be printed before the scoring step runs. `build_rerun_three_way.py` recomputes every verdict at that point, under the sampling-noise rule plus the per-competition variance floors of Section 3.2.

Table 4: The 20-competition benchmark, state as of 2026-08-17 (evening). my-agent’s local CV (OOF) is each lane’s own honest out-of-fold estimate, not leaderboard-validated (Section 5.1); its public/private leaderboard cells are TBD pending the scoring step. NVIDIA and AIDE columns are frozen private-leaderboard scores with percentile rank (PR) in parentheses. 15 of 20 lanes are complete (~17 hours of lane time), one is running, four are queued.

#	Competition	Metric	Local CV (OOF)	LB (pub/priv)	NVIDIA (PR)	AIDE (PR)	Lane status
1	s3e1	RMSE ↓	0.548862	TBD	0.55550 (93.2)	0.55552 (81.0)	done
2	s3e3	ROC-AUC ↑	0.841605	TBD	0.89537 (66.8)	0.86863 (26.4)	done
3	s3e5	QWK ↑	0.571980	TBD	0.56974 (75.7)	0.58138 (79.8)	done
4	s3e7	ROC-AUC ↑	0.901769	TBD	0.90458 (78.1)	0.90109 (67.1)	done
5	s3e9	RMSE ↓	12.082282	TBD	12.22913 (59.1)	12.30667 (44.5)	done
6	s3e11	RMSLE ↓	0.292886	TBD	0.29624 (66.6)	0.29445 (73.3)	done ^a
7	s3e14	MAE ↓	340.95936	TBD	330.71616 (89.2)	332.31556 (71.2)	done
8	s3e16	MAE ↓	1.334958	TBD	1.34001 (96.4)	1.34224 (91.4)	done
9	s3e19	SMAPE ↓	6.70585	TBD	48.26558 (41.4)	48.34131 (40.1)	done ^b
10	s4e11	Accuracy ↑	0.940810	TBD	0.94076 (52.8)	0.94056 (62.1)	done
11	s5e10	RMSE ↓	0.055971	TBD	0.05588 (59.4)	0.05579 (78.9)	done ^c
12	s6e1	Error ↓	TBD	TBD	8.73239 (69.5)	8.72191 (74.8)	running
13	s6e2	ROC-AUC ↑	TBD	TBD	0.95508 (63.0)	0.95516 (76.5)	queued
14	afsis	MCRMSE ↓	0.470503	TBD	0.69855 (18.8)	0.49861 (45.6)	done
15	cat-in-the-dat	ROC-AUC ↑	0.803272	TBD	0.77084 (32.6)	0.80217 (70.3)	done
16	conway	MAE ↓	0.106688	TBD	0.11189 (96.5)	0.11065 (97.9)	done
17	tps-aug-2022	ROC-AUC ↑	TBD	TBD	0.58730 (43.0)	0.59106 (60.5)	queued
18	tps-jan-2022	SMAPE ↓	TBD	TBD	4.63651 (76.4)	10.66638 (28.9)	queued
19	s5e1	MAPE ↓	5.84%	TBD	0.14594 (69.3)	0.17917 (48.1)	done ^d
20	tps-sep-2022	SMAPE ↓	TBD	TBD	5.48351 (80.3)	9.72095 (48.3)	queued
Median PR (range)			—	TBD	68.0 (18.8–96.5)	68.7 (26.4–97.9)	

^a First attempt exited without a submission; completed 08-17. ^b Clean re-run 08-17; first attempt discarded by the transcript audit (Section 4.3). ^c Audit incident resolved, score accepted (Section 4.3). ^d The lane’s cross-arm blend weights and global scale are fitted on the same OOF rows they are scored on, so its 5.84% is optimistic relative to a held-out estimate — the lane’s own summary records this caveat.

Table 5: Standing under the paired test: 60 duels, recomputed once the leaderboard scoring step runs. All cells await that step; the yardsticks’ rows are also TBD because their win rates aggregate over duels against my-agent.

Method	Wins	Losses	Win rate (excl. ties)
my-agent	TBD	TBD	TBD
NVIDIA	TBD	TBD	TBD
AIDE	TBD	TBD	TBD

5 Analysis: Development Findings That Shaped the Architecture

The two studies in this section ran no new competitions: they are validated readings of development-run artifacts — scores, logs, and champion code produced while the architecture was being built. They are reported not as current standings but as the findings that motivated two design decisions: submitting every benchmark verdict to the real leaderboard, and building the problem-identification layer of Section 2.2.

5.1 Local CV and the Leaderboard Disagree

Why. The attached per-competition specification reports (Appendix A) compare the agents on local CV, while the benchmark uses the actual leaderboard. If the two agree, CV is a reliable proxy and agent evaluation need not pay the cost of real submissions; if they disagree, every CV-based agent benchmark needs re-examination. The comparison is free — both datasets already exist.

Result: 9 of 13 competitions disagree (the comparison is limited to the 13 Playground competitions covered by the spec reports; CV verdicts, including their noise-floor tie calls, come from those reports’ scored comparisons, and leaderboard verdicts from the paired test, all on development-run lanes). The most extreme case, s3e19 (Table 6), fully reverses direction: CV says my-agent leads by 3.7 points; the leaderboard says it trails by 2.2. Other disagreement patterns include: CV declares a tie while the LB separates the sides (s3e7, s3e9, s3e11, s5e10, s6e1), and CV declares a winner while the LB leaves that agent’s duels undecided (s3e16).

Table 6: The most extreme CV–LB disagreement, s3e19 (SMAPE, lower is better; development runs). CV numbers are a matched-split diagnostic pair (same TimeSeriesSplit for both agents), not the master table’s champion CVs.

Basis	my-agent	NVIDIA	Verdict
Local CV (TimeSeriesSplit, same split for both)	10.02	13.72	my-agent by a wide margin
Private leaderboard	50.46	48.27	NVIDIA wins

Why this happens, and what follows. CV and LB measure different things — performance on resamples of the training data versus on a batch of never-seen data — and they decouple as soon as the test–train relationship changes. s3e19 is the extreme case: the matched-split CVs sit at 10–14 (AIDE’s own selected CV, 4.4, is the leaked shuffled-KFold number of Section 5.2) while the LB scores sit at 48–50, everyone misfiring together into the lower mode of a bimodal leaderboard (Section 5.2). The methodological implication is that agent benchmarks must submit to the real leaderboard: CV’s agreement rate here is only $4/13 = 31\%$ (95% CI 9–61%). The attached per-competition reports are therefore explicitly labeled CV-only, and the local CV column of Table 4 is quoted for progress, never for comparison.

5.2 Failure-Mode Analysis

Score tables cannot answer “why.” Reading the three agents’ development-run execution logs and champion code competition by competition revealed systematic method-level weaknesses — worth more to agent design than any ranking, and the direct motivation for the dossier layer.

AIDE: can diagnose, but no mechanism lets the diagnosis veto the score. On s3e16, at steps 16 and 18 AIDE ran nested CV on its own initiative, measured calibrated values of 1.3396 / 1.3412, and wrote in its own report that “the in-sample number is optimistic” — yet still selected the in-sample node; the private LB of 1.34224 confirms the honest estimate it discarded was the correct one. On s3e19, worse: mid-run, while fixing a LightGBM API issue, AIDE accidentally swapped its time-based validation for shuffled KFold, its metric instantly “improved” from 18.42 to 4.41, and nothing flagged the jump — on time-series

data, shuffling lets the model validate on rows from inside sequences it has already memorized — so it spent its remaining nine steps tuning against a leaked number. A controlled experiment that changes only the split measures the illusion at 4.5×. my-agent ran the same shuffled split in the same competition, but tagged it “diagnostic only” and never submitted it. The full step trajectory is in the companion case-studies document.

NVIDIA: votes do not equal quality. The method selects kernels by votes, but votes measure popularity and pedagogical value. On s3e3, a battle-tested solution with only 89 votes ([chunweishen/ps-s03e03-ensembling](#)) took the three-way best in development runs, while high-vote kernels are often tutorial-oriented rather than competition-oriented (in a getting-started competition outside this report, a 7,859-vote tutorial kernel left NVIDIA last of the three). Moreover, the s3e19 kernel it reproduced ([tumpanjawat/s3e19-course-eda-fe-lightgbm](#)) claims in its header “same idea as the 2nd-place solution”; NVIDIA faithfully reproduced its skeleton but missed the year-scaling step that decides the outcome (48.27 vs. the leaderboard top of 4.67) — nothing in the method ever checks the reproduced score against the source’s claimed rank.

A gap shared by all three: recognizing the need for external data. The s3e19 leaderboard splits into two clusters with almost nobody in between: of 1,174 teams, 265 score below 10 while 570 land in 40–60. In the development runs all three agents landed in the worse cluster (private SMAPEs in the 48–50 band). The teams in the better cluster did something none of the three agents did: they joined per-country GDP data. A correction we owe our own midterm: the midterm blamed the missing GDP join for the whole gap, but at matched budget the join is worth −6.0 SMAPE (Section 6.2) against a ~44-point gap to the leader — real, but small; the dominant obstacle is a full-future-year extrapolation that defeats every method in the comparison equally. There is a second lesson in how we got this wrong: our experience library already contained the correct diagnosis, but we retrieved and acted on a neighbouring entry that over-claimed — the library’s weak link is retrieval, not content (Section 8.5). Nor is this a task-type ceiling: on the same-type tps-jan-2022, my-agent proactively joined World Bank GDP-per-capita data (CV SMAPE 4.1793), and AIDE, using only official data, discovered calendar and Black-Friday features on its own (6.1551).

6 The Injection Layer: Controlled Experiments

Acting on the diagnosis of Section 5.2 — problem-level judgment is the shared gap of all three agents — the experiments here evaluate the injection layer of Section 2.2 under the same leaderboard discipline as the benchmark: real submissions, judged by the paired test of Section 3.2.

6.1 Problem-Identification Sweep (Dossier Layer)

Design. The dossier layer’s generality claim is “behaves correctly across the task distribution”: fire on tasks that genuinely need external data (sensitivity), stay silent on those that do not (specificity). We ran the dossier stage on all 18 original benchmark competitions — LLM reasoning over the problem statement, data schema, and task priors only; minutes per competition; no training.

Result: perfect separation. The layer fires on exactly the two country-panel time-series competitions — s3e19 and jan-2022, the two whose accepted solutions are known to require external covariates — and reports “no / unlikely” on all 16 iid-tabular, categorical, spectral, and structured-output competitions: sensitivity 2/2, specificity 16/16. (Two caveats keep this honest. The sensitivity half is partly in-sample: the task priors the dossier reads were distilled from this benchmark’s own failure analysis of exactly these two competitions, so the 16/16 silence is the informative half, and 2/2 carries a 95% CI reaching down to 16%. And the two firing-class competitions added to the baseline later also fire, but they were *selected* as firing-class, so the sweep claim rests on the original 18.) Two secondary behaviors: on s3e19 the

dossier surfaced a genuine conflict — a conservative setup-time `config.yaml` flag said external data was not allowed while the task prior’s evidence implied otherwise — and flagged it for verification rather than guessing (the official rules, Section 7.C, settle it in favor of permission); on the out-of-scope SIIM imaging competition it answers “no”, correct within the current whitelist, whose source classes do not yet include prior-competition archives.

6.2 What Upstream Injection Is Worth, and What Selects Its Form

Design. s3e19 is the benchmark’s canonical failure, so it is the causal test bed; tps-sep-2022 — six-country daily sales, outside the frozen baseline, dossier written before any of the findings below — is the held-out replication. For each competition we build arms whose processed feature tables are digit-identical to the baseline’s plus whatever the operators add, so a configuration that drops the new columns reproduces the committed baseline score exactly (verified: difference 0.0). Each arm is submitted for real public and private scores and judged by the paired test of Section 3.2.

Result 1a: external-data injection works, robustly. All four matched-budget comparisons in Table 7 beat the no-external baseline.

Table 7: Matched-configuration arms: identical pipeline, no search, no blending, so the operator is the only difference. SMAPE, lower is better. Every external arm beats its own baseline and passes the paired test of Section 3.2. s5e1, which joined the firing class later, skipped this matched-configuration stage — its pre-registered test ran searched arms only (Result 2b).

Arm	s3e19				tps-sep-2022 (held out)			
	CV	public	private	sig.	CV	public	private	sig.
no external data	10.148	52.492	54.506	—	11.348	23.828	25.476	—
join_feature	9.379	47.036	48.497	yes	11.624	21.973	23.390	yes
ratio_target	7.793	50.853	52.073	yes	11.807	21.209	24.091	yes

Against the baseline: on s3e19 the feature join gains 6.01 private SMAPE and the ratio target 2.43; on sep-2022, 2.09 and 1.39. The paired movements are small relative to the gaps (0.55, 0.79, 0.23, 1.23 respectively), so all four survive. A with-layer versus without-layer ablation — the full pipeline with the injection layer active, against the frozen pre-injection lane’s searched champion — also passes the paired test (with-layer private 48.243 vs. 50.455 without; $d_{\text{pub}}+3.389$, $d_{\text{priv}}+2.212$, movement 1.177), and places the with-layer champion numerically ahead of both frozen yardsticks on s3e19 (48.26558 and 48.34131, Table 4).

Result 1b: the ordering survives a real search budget, and search does not rescue the weaker form.

At an equal 46-node budget per arm on tps-sep-2022, the no-external arm reaches CV 11.347 / private 25.472; ratio_target 10.829 / 23.774; the gated variant (ratio paired with sample_weight on 2020) 10.922 / 23.973. Both external arms beat the searched baseline and pass the paired test (movements 0.499 and 0.577 against gaps of 1.698 and 1.499). The informative comparison is across protocols: 46 nodes of search on the *weaker* operator form (23.774) still lose to a *single unsearched configuration* of the stronger one (23.390; a cross-protocol comparison, not a paired verdict) — the choice of injection form is worth more than the entire search budget.

Result 2a: cross-validation cannot choose between the forms, and errs in opposite directions. On s3e19 CV ranks ratio_target best (7.793) while the leaderboard ranks it middle; on sep-2022 CV ranks *both* external forms *behind* the baseline while the leaderboard has both beating it significantly. A cross-validation whose folds all sit inside the training range cannot measure an operator whose purpose is extrapolation beyond it — and on sep-2022 the last folds land on 2020, the year the proportionality

diagnostic flags as broken (cross-country dispersion 0.466 against ~ 0.01 elsewhere), so the operator is scored twice on its worst case; we predicted the inversion from the fold boundaries before submitting. Nor is CV’s verdict stable in the budget: at one node it ranks `ratio_target` behind the baseline, at 46 nodes ahead; the leaderboard says “ahead” in both cases.

The diagnostic’s own prescribed remedy — down-weighting the broken 2020 to 0.3 — improves CV as designed (11.807 to 11.515) but is slightly worse on the leaderboard (private 24.142 vs. the ungated arm’s 24.091): CV misranks not only the operator choice but also the repair of a violated precondition. And a held-out-year protocol — the obvious cheap substitute, whose validation geometry matches the task — also fails: across three probes (s3e19 2021, sep-2022 2019 and 2020) it ranks `ratio_target` first every time, while the leaderboard ranks `join_feature` first on both competitions. Selecting the *form* of an upstream injection is therefore not currently possible from local signals, and the leaderboard cannot stand in for them in a live competition.

Result 2b: a pre-registered selection rule, killed by its own test (s5e1). After the three probes above failed, one candidate selector remained standing: the proportionality diagnostic, which had matched the outcome on both competitions — but only in hindsight. A fourth firing-class competition (playground-series-s5e1, “Forecasting Sticker Sales”), which the diagnostic read as the worst proportionality violation in the class, allowed a real test: before any model was fitted or anything submitted, we committed a prediction to the competition’s dossier and to version control — `ratio_target` loses to `join_feature` here, by a larger relative margin than on either predecessor — with the falsifier stated alongside (the diagnostic’s full reading of s5e1 and the panel-family details are in the companion case-studies document).

The falsifier fired. Three 45-node searched arms, identical except for the operator’s added columns, scored public/private MAPE 0.14280/0.15751 (baseline), 0.14193/0.15626 (`join_feature`), and 0.12057/**0.12417** (`ratio_target`) — the predicted loser won by a 20% relative margin, paired-test significant on all three pairwise comparisons, with no remedy applied for the violated years. Per the consequence pre-committed with the prediction, the proportionality diagnostic is demoted from a gate to a recorded note: it describes the data honestly but does not predict which operator form wins.

Three things follow. First, the injection claim itself *replicates at $n=3$* : across three competitions, all six external-data arms beat their no-external baselines, every one paired-significant, s5e1 by the largest margin yet. Second, the form-selection problem hardens rather than resolves: the form verdict is now split (`join_feature`, `join_feature`, `ratio_target`), and every local signal tested is 0-for-3 — CV picked `ratio/baseline/featurejoin` where the leaderboard said `featurejoin/featurejoin/ratio`, the held-out-year protocol misranked all three probes, and the proportionality diagnostic died on its first advance prediction. The operating rule is now: race both forms, always. Third, the two 1-year-horizon competitions went to `join_feature` and the one 3-year competition to `ratio_target`, so horizon length may be the real selector — but that is post-hoc pattern-matching, exactly the move the proportionality story was built on, and it earns belief only by pre-registration on the next firing competition. (jan-2022, the class’s fourth member, is not a clean test bed — its outcomes are already inside the prior library — but its record, a ratio-family decomposition beating a GBDT at a 1-year horizon, is counter-evidence the horizon story will have to explain.)

Result 2c: the full three-way on the new firing-class competitions. The controlled arms above isolate operators; they do not say how the three *agents* compare on this class. Both new firing-class competitions were therefore run by all three agents under benchmark conditions (these runs are what admitted s5e1 and sep-2022 into the 20-competition baseline of Section 4.2) — my-agent’s full pipeline with the knowledge library pinned, for each lane, to the last commit before any information about that lane’s own competition entered it (its own leaderboard verdicts would be an answer key; class-level knowledge from the sibling firing competitions, produced partly by this study’s own leaderboard probes, remained available — an asymmetry the cold-start reference methods do not share); AIDE and NVIDIA frozen exactly as in the main benchmark — and every submission scored on the real leaderboard. These are the injection study’s own scored lanes; the my-agent column of Table 4 is the frozen-architecture benchmark lane for the same

competitions, to be scored in the separate operator step (TBD in Table 4).

Table 8: Firing-class three-way, real leaderboard (both metrics ↓). NVIDIA’s public scores match its source kernels’ claimed LB digit-for-digit, verifying the faithful port. Verdicts per the paired test of Section 3.2.

Competition	Board	my-agent	NVIDIA	AIDE	Verdict
s5e1 (MAPE)	public	0.10236	0.05726	0.12573	
	private	0.12752	0.14594	0.17917	$M > A$; M vs N sign-flip
sep-2022 (SMAPE)	public	28.32738	4.77458	5.78998	
	private	28.54859	5.48351	9.72095	$N > A > M$

The private ranking reverses completely between two competitions of the same narrow class: $M > N > A$ on s5e1, $N > A > M$ on sep-2022. Sharper still, on s5e1 the paired test refuses to order my-agent and NVIDIA at all: the public half of the test set says NVIDIA by 0.045, the private half says my-agent by 0.018 — the evaluation-set caution of Section 8.3, that a ranking is a property of the method *and* the evaluation set, reproduced between the two disjoint halves of a single competition’s test split. The mechanism is visible in the degradation: the vote-selected kernel was developed against the public board it is tuned to ($0.057 \rightarrow 0.146$, +155%, on s5e1), while my-agent’s CV-driven pipeline degrades least (+24.6%) and takes the private board.

sep-2022 is the mirror image and NVIDIA’s textbook win: a mature public solution exists (a Ridge baseline, 277 votes), reproduction inherits it, and all three pairwise verdicts are significant. my-agent’s 28.5 there is another CV–LB decoupling entry: its champion bet a wrong 2021 regime from the pinned GDP prior with CV 4.67; the lane’s unchosen equal-share arm scores 16.25 private, submitted flagged as a diagnostic and not reported as an agent result. One further observation closes the loop with the injection results: the s5e1 kernel NVIDIA reproduced *itself* joins external GDP-per-capita data — the same covariate the dossier layer identifies, both plausibly drawing on this competition family’s shared folklore — so vote-selection inherits the injection for free exactly when a mature public solution exists, and when the public board misleads instead, the strategies separate.

Result 3: decomposing what remains, and correcting our own explanation. Diagnostic probes — global rescalings of the best arm, labelled as diagnostics and never reported as agent results — separate level from shape on s3e19: roughly 15.5 points of the residual are a yearly-level error and roughly 33 remain as shape, against a leaderboard top of 4.67. The needed 1.6 level multiplier is unlearnable in sample, external GDP data is worth 6.0 of the 44-point gap — the midterm correction recorded in Section 5.2 — and the experience library’s other candidate recipe, multiplicative panel decomposition, fails to transfer here despite an apparently satisfied precondition. The probe grid, the in-sample evidence on the multiplier, and the full transfer test are in the companion case-studies document.

What the residual means for the report’s central claim. The same pipeline scores 11.59 predicting 2021 from 2017–2020 and 48.497 predicting 2022 from 2017–2021 — a fourfold degradation from an identical procedure, which neither the level correction nor the structural recipe explains. That is Section 5.1’s “everyone misfires together” quantified: the final transition of a synthetic series can differ in kind from every transition inside the training window, and no in-sample protocol can anticipate it.

What the judgment layer proposes, and what actually consumes it. Across the 20 dossiers the layer proposes 241 ideas. Counting those whose operator family exists gives 73%; tracing which are actually *consumed* gives only 5% at first measurement, because the three operators supplying most of that headline were emitted into a key that no search driver read, while the dispatcher recorded them as realized (now 27%, after wiring a consumer and moving the target-free encodings into the data layer). A produced artifact with no consumer is indistinguishable from a working one in every log, score table and coverage

metric — and this instance occurred inside the very mechanism built to prevent silent truncation. **An architecture claim needs a consumer trace, not a vocabulary count.** Per-regime tables and audit scripts are in Appendix A.5.

7 Generalization: Mid-Complexity Three-Way

Design. The three agents run, each under its frozen method, on three competitions outside the tabular Playground family: us-patent-phrase-to-phrase-matching (NLP, text pairing), ventilator-pressure-prediction (physiological time series), and SIIM-ISIC melanoma (dermoscopic imaging) — machine fully dedicated per lane, 4–8h budgets, same isolation protocol. This tests whether the Playground conclusions (ranking reversal, CV–LB decoupling, failure modes) extend beyond the task family they were measured on. Version discipline: none of the three lanes ran the injection layer (Section 8.4). Search discipline: none of the three lanes used tree search — a node here is a full deep-learning fine-tune (30–90 minutes, versus seconds for a tabular GBDT refit), so the prescribed 25–80 node budget would cost 15–100+ GPU-hours against the 4–8h available. All three took the skill’s documented fallback for exactly this condition: one linear pass to a validated single model plus a blend, with the high-leverage moves (backbone choice, seed bagging, blend weights) taken directly.

Scoring. The mid-complexity competitions use MLE-bench [4] prepared splits and are graded offline against the held-out private split (us-patent is a code competition and ventilator/SIIM live test sets differ from the prepared splits, so Kaggle late submission is structurally unavailable); medal thresholds and the above-median flag come from the frozen leaderboard snapshot.

Table 9: Mid-complexity three-way, offline-graded against the held-out private split. Bold = three-way best; † = above the frozen leaderboard’s median.

Competition	Metric	my-agent	NVIDIA	AIDE
us-patent	Pearson r ↑	0.86658 † (silver)	0.86160† (bronze)	0.65821
ventilator	MAE ↓	0.16525	0.40093	0.34591
SIIM-ISIC	AUC ↑	0.93529 †	0.89822	0.88579

Results. my-agent takes the three-way best on all three and the only above-median score on SIIM — a clean sweep across three unrelated task families, so the result is about pipeline discipline rather than tabular-specific tuning. Offline grading yields one score per submission, so the paired public/private test of Section 3.2 does not apply here.

8 Discussion

8.1 Contributions

1. **An end-to-end competition agent whose knowledge enters at problem identification rather than at blending** — with the limits of that claim stated. The agent runs the full pipeline under LLM control with Auto-ML tools, an experience library, and a tree-search optimisation loop; the architectural claim this report argues for is the upstream dossier layer. It separates cleanly on the benchmark (fires on exactly 2 of 18, no false positives); in matched-configuration designs every external-data arm beats its own baseline significantly — six of six across three firing-class competitions — against the near-null effect the same pipeline measured at the blend stage (different competitions and metric scales; the contrast is qualitative, not a same-unit ratio); and on the benchmark’s canonical failure the with-layer/without-layer ablation is paired-test significant. What we cannot do is choose which *form* the

injection should take: every local signal tested is 0-for-3, including a diagnostic we believed enough to pre-register a prediction on — which its own named falsifier then killed.

2. **The agent’s placement against two frozen, methodologically distinct yardsticks, with the adjudication rule committed before the scores exist.** The 20-competition benchmark lanes are run, their submissions frozen, and their leaderboard standings TBD pending the credentialed scoring step; when scored, all 60 duels are adjudicated by the paired test and per-competition variance floors of Section 3.2. What is scored already: the three-way best on all three mid-complexity competitions (NLP, physiological time series, medical imaging), and the firing-class three-way of Result 2c, which characterises *where* each method’s advantage lives — reproduction inherits mature public solutions for free, and the strategies separate when the public board misleads (Section 5.2).
3. **Two cautions that constrain how any such claim may be stated.** A ranking is a property of method $\times$ evaluation set — in development runs the winner flipped between two reasonable competition sets, and on s5e1 between the two halves of a single test split (Section 8.3) — and local cross-validation is not a proxy for the leaderboard, disagreeing in 9 of 13 development comparisons and fully reversing direction on s3e19 (Section 5.1). Both are demonstrated rather than asserted, and both apply to our own agent first.
4. **Measurement apparatus, offered for reuse.** A paired significance test that costs no extra experiments — Kaggle’s own public/private split supplies the error scale — together with a direct measurement of between-run agent variance, and an execution discipline (credential-free lanes, manifest-frozen workspaces, per-lane transcript audit) that has already discarded one contaminated lane whose tainted number looked better than its clean replacement (Section 4.3). This is infrastructure rather than a finding, but without it none of the numbers above would be readable.

8.2 Comparing the Three Methods

The differences lie not in tuning skill but in where knowledge comes from and how the next step is decided (Table 10).

Table 10: Design comparison of the three methods.

Aspect	my-agent	AIDE	NVIDIA
Knowledge source	Cross-competition experience library (accumulating)	None (from zero each time)	Public kernels fetched fresh each time
Search strategy	Six-stage pipeline + tree search (25–80 nodes)	Tree-shaped trial and error, 20 steps	No search; faithful reproduction
Basis for next step	Experience-library priors + OOF diagnostics	Previous step’s error or score	Highest-voted kernel
Validation rigor	Split chosen by task type; diagnostics tagged and excluded	No concept of split legality	Inherits the kernel’s split
External data	Injection layer (typed operators, whitelisted sources)	Never considered	Inherited from the kernel

Respective strengths, at the mechanism level.

- **NVIDIA’s strength is standing on shoulders, not its own reasoning.** Where a mature, high-quality public solution exists, it skips the entire exploration process and directly harvests hundreds of community person-hours — sep-2022, with its 277-vote Ridge baseline, is the textbook case (Result 2c).
- **AIDE’s strength is search depth.** On s3e11 its edge came from a bug-fix node that accidentally enabled native categorical handling — massive trial and error stumbles into combinations a person would not think of. It needs no external knowledge to operate.

- **my-agent’s strength is validation rigor and knowledge accumulation.** Facing the same shuffled-KFold trap on s3e19 (Section 5.2), AIDE misused it for its official submission while my-agent quarantined it — a distinction that never shows up in a single score, yet determines reliability on unseen tasks.

Respective limitations.

- **NVIDIA’s bottleneck is its selection signal.** Votes measure popularity, not quality (Section 5.2); it never checks the reproduced score against the source’s claimed rank, and when no public solution exists, the method has nothing to work with — and its one-shot execution is all-or-nothing: when process discipline lapses, there is no fallback, where iterative search always holds a submittable solution (engineering log, ventilator reconstruction).
- **AIDE’s bottleneck is the missing mechanism to let diagnoses veto scores.** It can produce correct diagnoses yet still choose the optimistic node (s3e16), and on s3e19 its own metric improved 4× in one step without raising any alarm (Section 5.2). It optimizes the displayed number, not true performance.
- **my-agent’s bottleneck was judgment at the problem-identification level** — the gap the injection layer was built to close, and the reason its controlled experiments (Sections 6.1 and 6.2) carry the report’s thesis. What remains open within the layer is form selection (0-for-3) and the unmeasured value of the dossier on competitions where it stays silent.

Shared limitation. On s3e19 all three development runs landed in the lower mode of the bimodal leaderboard, none recognizing the need for GDP data — a per-competition miss, not a task-type ceiling (Section 5.2). Section 6.2 sharpens this: the join is worth 6.0 SMAPE, but the shared shortfall is mostly a one-year extrapolation no method in the comparison addresses.

8.3 A Ranking Is Not a Property of the Method, but of Method × Evaluation Set

The most important methodological lesson of the development runs: same agents, same test — swap the competition set and the paired-test winner flipped between the 14-competition Playground subset and the broader 20-competition set. (Those standings belong to superseded development lanes and are not reprinted here; the benchmark’s standings are TBD pending scoring. The mechanism, however, is a property of the methods and survives.) Three of the six non-Playground competitions are old or non-mainstream (afsis 2014, conway 2014, cat-in-the-dat 2019): their public kernels are either stale or tutorial-grade, so NVIDIA’s “reproduce the top-voted kernel” loses the mature solution ecosystem it enjoys in the Playground series — its frozen PR drops to 18.8 (afsis) and 32.6 (cat-in-the-dat, Table 4). The exception proves the mechanism: sep-2022, the one non-Playground competition with a mature public solution (a 277-vote Ridge baseline), is NVIDIA’s cleanest win in the set (Result 2c). Result 2c also reproduces the caution inside a single competition: on s5e1 the public and private halves of one test split order two agents oppositely.

Two observations survive from the development-run reading. First, absolute-position metrics (median PR) ordered the three identically on both evaluation sets while the paired win rate reversed — relative verdicts binarize many near-tie gaps and are therefore highly sensitive to sample composition. Second, “which agent is stronger” is an incomplete question; the correct question is “stronger on what distribution of competitions”: with mature public solutions, reproduction dominates; on obscure, old, build-your-own-pipeline tasks, autonomous methods dominate. An agent benchmark’s conclusion must be stated together with the composition of its evaluation set, and should report both an absolute metric (PR) and a relative one (paired win rate).

8.4 Limitations of the Experimental Design

- **The significance test covers only test-set sampling noise**, not between-run agent variance. That variance is measured for my-agent and AIDE on four competitions (Section 3.2) and enters the combined error model as a per-competition floor; NVIDIA’s term remains unmeasured.
- **The duels are not independent trials**. Each competition contributes three duels that are deterministic functions of six shared scores, and each submission enters two duels, so duel-iid intervals are anti-conservative and the effective sample size is closer to 20 competitions than to 60 duels; interval estimates on the scored standings must use competition-level cluster bootstrap. A further asymmetry touches every NVIDIA duel: reproduced kernels were tuned by their authors against the realized public leaderboard, so NVIDIA’s public score is not an unbiased estimate of its private one. In the development lanes the median public-to-private degradation was small for all three (+0.5 to +0.8%), but NVIDIA’s distribution sat highest — IQR [+0.3, +3.6]% — and held the extreme (+155%, s5e1). The paired test’s exchangeability premise is weakest exactly where the reproduction method is strongest.
- **Budgets are per-method conventions, not compute-matched**. AIDE runs its default 20 steps, my-agent 22–80 evaluated nodes plus a linear pass, NVIDIA minutes of reproduction; each method runs “as shipped,” so the comparison is methods-with-their-budgets, not methods at equal compute.
- **The baseline expansion was post hoc**. The two firing-class competitions’ leaderboards had been consulted for the injection experiments before their three-way lanes ran; in the development runs the addition moved every rate, so the 18-vs-20 composition is disclosed rather than hidden.
- **The asymmetry created by the experience library is disclosed, not eliminated**. my-agent carries cross-competition memory; the other two cold-start — a design difference rather than a flaw, but it colors the interpretation of the conclusions.
- **The sample is small, and Section 8.3 shows the ranking is sensitive to set composition**. With 20 competitions and 60 duels, adding or removing a single competition can move any win rate by one to two percentage points — the two firing-class additions themselves moved every development-run rate.
- **The mid-complexity lanes ran without the injection layer and without tree search** (Section 7), and their offline grading yields one score per submission, outside the paired test’s reach; their sweep is therefore a raw-score result, not a paired-verdict one.

8.5 Open Problems

Ordered by how much they would change this report’s conclusions.

NVIDIA’s between-run variance is unmeasured. The variance measurement of Section 3.2 covers my-agent and AIDE on four competitions; on those competitions, every development-lane verdict that survived the combined error model involves NVIDIA — the one agent whose variance term is missing. Widening the sample beyond the four competitions measured is the cheapest way to harden every verdict the benchmark will produce.

Operator form cannot be selected from any local signal we have tested — 0-for-3, with one selector killed by pre-registration. Section 6.2 details the misrankings; the split form verdict means no constant answer exists to fall back on. The one surviving candidate selector, horizon length, is post-hoc; testing it requires pre-registering it on the next firing-class competition before its leaderboard is consulted (the binding protocol is committed as the H-HORIZON pre-registration). This remains the sharpest open question here, because the form choice was worth more than the entire search budget on one competition and a 20% relative margin on another.

The firing class is small. The dossier fires on 2 of 18 baseline competitions, so the controlled evidence rests on two clean test points plus one held-out replication. Extending the class is worth more than re-running the 16 competitions where the layer stays silent: a re-run there changes nothing the layer does and instead re-rolls 48 of the 60 duels with the run-to-run variance measured in Section 3.2.

The experience library’s weak link is retrieval, not content. While decomposing the s3e19 residual we re-derived a result the library already held, because nothing forces a query before an experiment; a neighbouring entry meanwhile over-claimed and has since been corrected against leaderboard evidence. The next layer worth building is a retrieval gate rather than more stored knowledge.

Smaller items. Whitelisting further source classes (prior-competition archives for the SIIM class, public text corpora for us-patent, both identified by the mid-complexity applicability analysis); an automatic split-legality check triggered by a single-step metric improvement beyond a threshold (AIDE’s two failures are the motivating cases); reproduce-then-verify against the source’s claimed rank, the concrete fix for NVIDIA’s vote bottleneck; and a direct ablation of the experience library.

9 Conclusion

This study set out to build an end-to-end competition agent and to find out honestly what its architecture is worth. The architectural claim is about placement: measuring our own first design showed knowledge injected downstream at the blend stage to be worth $\sim 10^{-5}$, so the architecture injects it upstream, at problem identification — a dossier layer that fires on exactly the 2 of 18 baseline competitions needing external data and stays silent on the other 16, an external-data module whose typed, leakage-guarded operators beat their matched baselines in all six controlled arms across three firing-class competitions, every one paired-significant, and a with-layer/without-layer ablation that passes the paired test on the benchmark’s canonical failure. Holding that layer to the same discipline also returns the negative result that sharpens it: no local validation protocol can choose the *form* the injection should take — 0-for-3, with the one selector that survived hindsight killed by its first pre-registered prediction, a failure we count as the method working, since a benchmark that can kill its own hypotheses is the thing this study set out to build — and a consumer trace showed that only 5% of the layer’s proposals were actually being applied at first measurement (Section 6.2). To place the agent, all 20 benchmark competitions ran under the frozen architecture in credential-free, transcript-audited lanes — an audit that proved its worth by discarding a contaminated lane whose tainted local score was better than its clean replacement’s — and on the three mid-complexity competitions outside the tabular family, my-agent takes the three-way best on all three. The benchmark standings themselves are TBD: the lanes’ submissions exist, the scoring step is a separate credentialed operation, and when it runs, every one of the 60 duels will be adjudicated by the error model stated in advance — the paired public/private test plus measured per-competition run-to-run variance. That restraint is the point. In this study’s own development runs, local cross-validation contradicted the real leaderboard in 9 of 13 comparable competitions, the ranking’s winner flipped between two reasonable evaluation sets, and on one competition it flipped between the public and private halves of a single test split; a capability claim about an agent, ours first of all, is therefore only as meaningful as the error model and evaluation set stated alongside it. The arc of the internship is a single loop closed: build the agent, benchmark it honestly, read its failures, fix the layer they identify, and re-verify under the same discipline that found the gap.

A Experimental Procedure and Per-Competition Specification Reports

A.1 Per-Competition ML Specification Reports (Attached)

Full specifications for every agent on every competition, written to the five sections of Tso-Jung Yen’s ML specification framework (`ml_pipeline_modules`) — Data / Models & Architecture / Training /

Inference / Evaluation & Benchmarking — per competition: 15 reports for my-agent ([kaggle/docs/ml_specs/](#), MD + PDF) and 17 for AIDE ([aideml-runs/docs/ml_specs_aide/](#), MD + PDF).

Every number in these reports is extracted deterministically by `collect.py/collect_aide_facts.py` from that competition’s structured records (`facts.json`, `eda_summary.json`, `experiments_tree_v3.json`, `journal.json`), never generated by an LLM; unavailable fields are marked “not recorded” rather than left blank or estimated. The my-agent report set additionally ships `RUBRIC.md`, an 8-item plan-alignment checklist verified per competition (all 15 pass 8/8). The reports are CV-only by construction (Section 5.1); their conclusions must not be mixed with leaderboard verdicts.

This appendix does not repeat per-competition content; below are only the cross-competition rules and the execution environment that individual reports cannot cover.

A.2 Cross-Competition Rule: Split Strategy Is Determined by Task Type

This is the single highest-impact design item in the study, visible only through cross-competition comparison (Table 11).

Table 11: Validation-split policy by task type.

Task type	Required split	Rationale
Time-series extrapolation (test set later than training period)	Time-based split (<code>year < y</code> vs <code>year == y</code>) or <code>TimeSeriesSplit</code>	Each validation block must be strictly later than its training window, mirroring the true train–test gap
Text pairing (same anchor in many rows)	<code>GroupKFold</code> by anchor	Otherwise one anchor spans train and validation, and the model memorizes anchors instead of semantic relations
Generic tabular (continuous i.i.d. target)	Shuffled <code>KFold</code> / <code>StratifiedKFold</code>	No temporal or group structure

Shuffled `KFold` on time-series data is strictly forbidden: on s3e19 this one error cost AIDE a 4.5× optimism bias, while my-agent quarantined the same split as diagnostic (Section 5.2).

A.3 Cross-Competition Rule: DL-to-GBDT Field Mapping

The specification framework contains several deep-learning-specific fields; this study is GBDT-centric, and all per-competition reports follow the mapping in Table 12.

Table 12: Mapping of deep-learning framework fields to their GBDT counterparts.

Framework field	GBDT counterpart
Learning Rate	Boosting shrinkage (<code>learning_rate</code>)
Learning Rate Scheduler	N/A — convergence governed by CV early stopping, not a schedule
Batch Size	N/A — full-dataset histogram boosting, not mini-batches
Model Complexity	Expressed as “number of trees × leaves,” not dense parameter count
Transfer Learning	N/A (no pretrained weights); analogue: prior injection from the experience library
Data Augmentation	N/A (tabular); analogue: feature engineering and seed bagging

A.4 Execution Environment and Reproducibility

Isolation and lane discipline are covered in Section 3.1; Table 13 records only the environment items not stated there.

Table 13: Execution environment.

Item	Detail
Hardware	ARM64 Linux (Ubuntu 24.04), NVIDIA GB10 GPU
Environment management	uv + uv.lock (frozen versions); Python 3.13
Monitoring	bench-watchdog.timer, checking every 15 minutes for (a) machine idle with work pending, (b) process alive but no output for 30 minutes, (c) individual driver death
Determinism	Fixed seeds; LightGBM with deterministic=true, force_row_wise=true, fixed num_threads — necessary for cross-process reproducibility
Per-step memory cap	32 GB (exceeding it marks the node buggy; a single step once consumed 76 GB and stalled the machine)
Output isolation	my-agent kaggle/competitions/; AIDE aideml-runs/; NVIDIA nvidia-kaggle-runs/

A.5 Companion Documents

Two companion documents carry material that answers different questions than this report does, each written for a different reader. *Building the Injection Layer: An Engineering Log* asks what broke while building the machinery, for someone rebuilding a similar system. *Per-Competition Case Studies* carries the single-competition forensic depth — the CV-leaderboard reversal mechanics, AIDE’s step-by-step split switch, the two experience-library entries behind the midterm correction, the s5e1 setting behind the pre-registered prediction, and the s3e19 residual decomposition in full — for a reader who wants the evidence behind the condensed claims of Sections 5.1, 5.2, and 6.2.

The engineering log contains: the per-regime coverage tables behind the consumer-trace finding of Section 6.2, with the audit scripts that reproduce them; the isolation mechanisms summarised in Table 1, together with the record of every occasion one of them failed, what it cost, and how it was caught; the minute-by-minute reconstruction of the voided ventilator attempt; and the nine defects found in a single day’s audit of the injection layer, six from the audit itself and three from verifying the fixes for the first six.

They share a document because they share a failure shape: a produced artifact with no consumer is indistinguishable from a working one — the transferable claim of Section 6.2; the log is the evidence behind it.

A.6 Tree-Search Size per Competition (my-agent)

Table 14: Tree-search node counts per competition (my-agent). The budget machine’s default cap is 60 with per-competition overrides (afsis 80); plateaus stop several runs below 25. tps-sep-2022’s lane finished on the linear protocol without entering tree search.

Competition	Nodes	Competition	Nodes	Competition	Nodes
s3e1	22	s3e14	62	s5e10	50
s3e3	22	s3e16	24	s6e1	48
s3e5	23	s3e19	22	s6e2	60
s3e7	63	s4e11	60	afsis	80
s3e9	26	cat-in-the-dat	65	conway	25
s3e11	24	s5e1	60	tps-sep-2022	—
tps-aug-2022	60	tps-jan-2022	60		

B Code and Skill Packaging

All my-agent code and benchmark artifacts are archived in the project repository github.com/930731haohao-afk/kaggle-agent (private; access granted on request). File paths in the tables below link to the corresponding files in that repository; `ai_agents/-` level infrastructure scripts are mirrored under `benchmark_infra/`, and cross-agent result artifacts under `benchmark_results/`.

B.1 my-agent Skill Definition

Table 15: Skill definition files.

File	Purpose
kaggle/.claude/skills/kaggle-agent/SKILL.md	Main skill definition: six-stage pipeline, tree-search switching condition, experience-library query timing
.../references/01_setup.md – 06_submission.md	Detailed per-stage instructions
.../references/07_tree_search.md	Tree-search protocol (preferred optimization mode)
kaggle/tree_search/harness_v3.py	Tree-search harness
kaggle/knowledge/experience.md	Cross-competition experience library (every entry carries an evidence field)
kaggle/knowledge/idea_bank.md	Pool of unvalidated ideas

B.2 Experiment Execution Scripts

Table 16: Experiment execution scripts.

File	Purpose
ai_agents/bench-comps/build_isolated_roots.py	Build manifest-validated isolated data roots; officialness judged via the Kaggle API, incl. members of official archives
ai_agents/lane_lock.sh	Lane mutex (atomic directory creation, reclamation on holder death, bounded waiting)
ai_agents/bench_watchdog.sh	Dual-signal monitoring: machine level (alive + advancing) and per driver
ai_agents/kaggle/run_myagent_headless.sh	Unattended my-agent execution (one headless session per competition)
ai_agents/aideml/run_comp.py	Single-competition AIDE run (supports AIDE_COMP_ROOT)
ai_agents/aideml/rerun_20steps.py	Batch AIDE runs (20-step budget)
ai_agents/aideml/collect_aide_facts.py	Deterministic fact extraction from AIDE journals
ai_agents/nvidia-kaggle-runs/run_ready_reproductions.py	NVIDIA reproduction runner (only scripts passing smoke tests)

B.3 Data and Results

Table 17: Data and result artifacts.

Item	Path
Three-way score master table	ai_agents/aideml-runs/three_way_scores.csv
Full three-way comparison report	ai_agents/aideml-runs/THREE_WAY_REPORT.md (with PDF)
AIDE per-competition ML spec reports (17)	ai_agents/aideml-runs/docs/ml_specs_aide/
my-agent per-competition ML spec reports (15)	ai_agents/kaggle/docs/ml_specs/
Coverage status table	ai_agents/BENCHMARK_COVERAGE.md
Benchmark-run manifest	kaggle/docs/rerun_manifest.json
Firing-class lane log (Result 2c runs)	benchmark_results/FIRING_CLASS_LANES.md
Per-competition experiment logs	kaggle/competitions/<comp>/experiments.json , experiments_tree_v3.json

B.4 Injection-Layer Components

Table 18: Injection-layer components (all in the project repository).

File	Purpose
<code>.../references/00_problem_dossier.md</code>	Stage 0.5 instructions: dossier schema, typed-operator requirement, forbidden inputs, downstream consumption
<code>knowledge/task_priors.md</code>	Evidence-backed task-level priors ([TASK-*) + external-data whitelist with leakage rules
<code>knowledge/injection_operators.md</code>	The contract between the judgment and execution layers: 10 typed operators, parameter schemas, the evidence for each, and the conditions under which each must not be used
<code>external_data/sources.py</code>	Whitelisted fetchers (11 World Bank indicators, ECB FX, OWID COVID, holidays, ISO), snapshot-recorded atomic caches
<code>external_data/join.py</code>	Leakage-guarded year/month/day joins; declared time units; row-order preservation; audit logs
<code>external_data/apply.py</code>	Operator dispatcher: realizes data-level operators, emits config for config-only ones, writes the coverage ledger, runs the proportionality and covariate-volatility preconditions
<code>external_data/validate_dossier.py</code>	Emission-side conformance check (schema, whitelist, provenance); non-zero exit gates a run
<code>external_data/coverage_audit.py</code>	Computes the coverage split behind Section 6.2 (tables in the engineering log)
<code>external_data/selftest.py</code>	14-section execution-side test suite incl. direct unit test of the leakage invariant
<code>tree_search/make_v5_arm.py</code>	Competition-agnostic arm builder: digit-exact variant workspaces + generated evaluators
<code>tree_search/run_arm_search.py</code>	Variant-lane race driver (harness_v3 budget machine)
<code>tree_search/holdout_year_probe.py</code>	Held-out-year form-selection probe (the negative result of Section 6.2)
<code>tree_search/structural_probe.py</code>	Structural-decomposition probe: tests whether the s3e19 residual is a shape error a GBDT cannot express
<code>tree_search/eval_support.py</code>	Per-fold target encoding, sparse-linear family, runtime-attestation writer (2026-07-31)
<code>tree_search/seed_from_ledger.py</code>	Translates emitted node configs into real seed nodes; writes <code>injection_consumed.json</code> (the consumer trace)
<code>external_data/verify_advisory.py</code>	Advisory verification: static parameter check upgraded by runtime attestation (<code>evaluator_attestation.json</code>)
<code>docs/preregistrations/horizon_length_form_selector.md</code>	H-HORIZON pre-registration: binding protocol on the next firing-class competition
<code>competitions/*/dossier.json</code>	20 dossiers from the sweep (Section 6.1)

B.5 Minimal Steps to Reproduce This Study

0) All paths repo-relative (the `ai_agents/-level` originals map per the mirror
note at the top of this section)

```

# 1) Build isolated data roots (per official Kaggle manifest)
python benchmark_infra/build_isolated_roots.py --write <comp>

# 2) my-agent, one competition
cd ai_agents/kaggle && claude -p "<see run_myagent_headless.sh PROMPT>" \
    --dangerously-skip-permissions

# 3) AIDE, one competition (20 steps)
AIDE_COMP_ROOT=<data-root> \
    python benchmark_infra/run_comp.py <comp> --shim --steps 20

# 4) NVIDIA, one competition
python benchmark_infra/run_ready_reproductions.py    # generates+runs per-comp reproduce scri

# 5) Submit and retrieve public/private scores
KAGGLE_API_TOKEN=$(cat ~/.kaggle/huang_token) \
    kaggle competitions submit -c <comp> -f submission.csv -m "<msg>"

```

Environment: ARM64 Linux (Ubuntu 24.04), NVIDIA GB10 GPU, uv-managed Python 3.13, versions frozen in `uv.lock`.

Every number in this report is traceable to the structured records listed above. This report will be updated when the leaderboard scoring step runs and as the items in Section 8.5 complete.

References

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- [4] Chan, J. S., et al. MLE-bench: Evaluating Machine Learning Agents on Machine Learning Engineering. arXiv:2410.07095 (2024).